

diag_03 — Servo Path Validation

Layer 3 of the MMAV bench suite · diag_03_servo_1.ino · protractor + fixed airframe · 115200 baud

What it validates

Exercises the full command path of the flight firmware so you can confirm each stage against the real surfaces:

```
raw cmd (incidence°) → CMD_LPF → servo = NEUTRAL[side] +  
REVSIGN[side]·GAIN[side]·incidence → rate limit (240°/s) → μs → servo
```

The onboard IMU is on the fuselage and cannot see the wings, so this test does not self-measure surface angle. The static modes drive a clean, held command and print exactly what they commanded (incidence, servo°, PWM); **you** read the surface angle against the fuselage with a protractor and pair commanded-vs-measured. Each point holds until you press a key. The step modes log the firmware command path at 100 Hz (firmware lag only; physical servo/linkage lag needs high-speed video of the surface).

Before you start

1. **Fix the airframe** and clear the surfaces — the wings **will** move.
2. Connect the **battery / BEC** to the servo rail. Servos do **not** move on USB power alone, so this test needs the battery (unlike the Layer 1 timing sweep).
3. Flash `diag_03_servo_1.ino`, open Serial Monitor at **115200 baud**.
4. Have a **protractor** ready; measure the wing chord angle relative to the fuselage. The board does not need to be level (no IMU/SD used).

Current calibration (printed on every menu / h)

Parameter	Left	Right
NEUTRAL (raw servo deg for 0° incidence)	103.5	90.0
GAIN (servo deg per incidence deg)	1.00	1.00
REVERSE	true	false

Global: neutral 90°, command limit ±60°, rate limit 240°/s, CMD_LPF $\alpha=0.55$, servo pins L=D4 / R=D3, travel 1000–2000 μs over 0–180°.

Command reference

Key	What it drives	What you do
n	Holds both wings at 0° incidence (neutral).	Check both surfaces sit at true neutral.
q	Quick smoke: neutral, then ±10° on L, R, both (auto-holds ~1 s each).	Watch each surface moves the right way / amount.
l	Left-only static sweep through the setpoints; right stays neutral. Holds each point.	Protractor-read both surfaces; record commanded vs measured → left GAIN & linearity.
r	Right-only static sweep (left neutral).	Same, for the right side.

b	Both-wings symmetric sweep (pitch-axis geometry).	Read both surfaces; checks symmetric / pitch response.
s	Sign check: $\pm$ roll and $\pm$ pitch cases; prints the expected direction. Mix: $\text{leftCmd}=\text{uPitch}+\text{uRoll}$, $\text{rightCmd}=\text{uPitch}-\text{uRoll}$.	Confirm physical direction matches your sign convention.
c	Clamp check: requests $\pm 80^\circ$, $\pm 100^\circ$ (beyond the $\pm 60^\circ$ limit).	Confirm the surface does not move past the $\pm 60^\circ$ clamp.
p	Step response, symmetric (L+, R+): 15° step, logs command path at 100 Hz.	Read t10/t50/t90 and rate-limit saturation (firmware lag).
P	Step response, differential (L+, R-).	Same, for a roll (differential) step.
m	Travel-limit ramp: L+, L-, R+, R- in 2° steps up to a 70° ceiling, holding each.	Press a key at the first resistance / sound change. Records safe range.
j	Jog NEUTRAL finder: writes a raw servo angle per side (no gain / reverse / rate limit).	Level each wing to 0° ; read off the per-side NEUTRAL to hardcode.
h	Reprint the menu (shows current calibration).	—

Static modes print one line per point: `tag | L inc=· servo· pwm· | R inc=· servo· pwm·`

Jog mode (j) sub-controls

- `l / r` — select side
- `+ / -` — $\pm 0.5^\circ$ · `] / [` — $\pm 5^\circ$
- `p` — print current angles · `x` — exit (prints the NEUTRAL values to hardcode)
- Level the wing chord to 0° vs fuselage, then copy the printed value into `SERVO_LEFT/RIGHT_NEUTRAL_DEG` and re-flash.

Recommended order

1. `j` — find / confirm per-side NEUTRAL (level each wing).
2. `l` then `r` — per-side GAIN & linearity over the sweep.
3. `b` — symmetric (pitch) geometry.
4. `s` — sign / mixing direction.
5. `c` — command clamp holds.
6. `p / P` — firmware step lag.
7. `m` — safe mechanical travel limit (last: gentle).

CAUTION (m travel-limit ramp): the goal is the safe mechanical range, not maximum throw. Check the horn and linkage first, and press a key at the first sign of resistance or a change in servo sound. Never drive the surface to the ceiling.